

Manuscript

Layer-1

稿件草稿 HTML 版（基于 docs/manuscript_draft.md）

Self-contained academic HTML with embedded SciencePlots figures (base64).

Literature-anchored Layer-1 surrogate for exploratory screening of indirect mitral annuloplasty

Working short title: FMR IMA Layer-1 exploratory surrogate (software / methods)

Venue class: computational biomechanics / research-software methods paper (CMBBE / MedEngPhys / RSOS-class) — not a clinical decision trial.

One-sentence argument: We provide a transparent, literature-anchored algebraic surrogate that reproduces selected Galili peak-systole anchors, supports Dryad-aware provenance, reports true fold-wise response-model cross-validation (and a separate blend-off diagnostic), and ranks exploratory IMA-CS / IMA-AP scenarios under explicit geometric and η assumption-prior uncertainty — without claiming new LHHM FEA, clinical recommendation, or patient-level external validation.

Data / reproduce: `python run_pipeline.py --seed 42 --paper · python tools/import_galili_dryad.py --from-fixture --process · python -m analysis.fit_response_model --write · python tools/loo_evaluate.py --write`

Verification levels: Level 0 = calibrated to / reproduced selected Galili cases; Level 1 = exploratory planning map + uncertainty-aware scenario ranker + fold-wise response diagnostic. Level 2 patient-specific LHHM/FSI is out of scope.

Abstract

Background. Computational IMA studies (Galili et al., *R. Soc. Open Sci.* 2022; doi:10.1098/rsos.211464; title: *Numerical biomechanics modelling of indirect mitral annuloplasty treatments for functional mitral regurgitation*) report suture/bridge shortening with peak-systolic geometry and leakage. Peak-systolic IMA-AP 50% yields AP **15.9 mm** (near-direct AP effect), not undeformed diastole **34.4 mm**. Device classes differ: MAVERIC evaluates **ARTO** (CS-IAS / IMA-AP-class); Carillon / REDUCE-FMR are IMA-CS-class.

Methods. We implement a reproducible Python Layer-1 surrogate: phenomenological mechanics plus a literature-calibrated leakage proxy. Dryad (doi:10.5061/dryad.bzkh1899d) supplies primary provenance for `contact_fraction` and SPH particle counts; scalar AP/ROA/leakage remain table-backed unless derivable. We fit fold-wise response models `f_ROA( $\Delta$  AP, device family, annular reduction, contact_fraction)` and `f_leak(ROA, coaptation proxy, contact, device mechanism)` under true leave-one-case-out. A discrete scenario ranker minimizes leakage-proxy regurgitation subject to an AP ceiling,

an illustrative NiTi engineering screen, and a Rottländer CS–LCx risk-screening threshold, reporting device-candidate P(feasible), independent η_{AP}/η_{CS} ranking stability, and family-separated Pareto frontiers. Dual-suture commissural factor is an exploratory hypothesis parameter (sensitivity 0.25/0.5/0.75/1.0). Ranking is assumption-driven exploratory analysis, not validated prediction.

Results (seed=42). Fold-wise response-model CV on the held-out Galili subset improves ROA MAE from rule-based blend-off **50.18 mm²** to about **6.4 mm²** and leak MAE from **1.445** to about **0.25** percentage points; IMA-AP 70% absolute ROA/leak errors fall from **94.17 mm² / 5.455 pp** to about **6.6 mm² / 0.20 pp** (internal engineering targets, not medical validation). The fixed rule-based surrogate still **substantially overestimates AP70 leak magnitude** ($\approx 2.46\%$ vs 0.13%) while only capturing a qualitative non-monotonic tendency — not an unqualified reproduction claim. Under the planning map the ranker evaluates **36** total points (**35** device candidates; pathology is not a candidate) and retains **30** feasible designs (**p_feasible_device_candidates ≈ 0.857**). Under the nominal dual-suture hypothesis, dual AP60 ranked first (leakage delta vs matched single ~ 0.0018 pp); ranking is assumption-sensitive.

Conclusions. A literature-anchored low-order surrogate can support **exploratory screening** of IMA strategy settings with inspectable assumptions, honest held-out reporting, and uncertainty language. It is not a preoperative clinical decision system and must not equate physics % with clinical regurgitant volume.

Keywords: functional mitral regurgitation; indirect mitral annuloplasty; surrogate model; exploratory screening; research software; Layer-1

1. Introduction

Functional mitral regurgitation (FMR) reflects ventricular remodeling, annular dilatation, and impaired leaflet coaptation. Indirect mitral annuloplasty (IMA) concepts remodel annular geometry without direct leaflet repair. **IMA-CS** (Carillon-class) and **IMA-AP** (CS–IAS; ARTO / MAVERIC mechanism class) are not mechanically interchangeable.

Galili et al. (2022) compared generic IMA-CS and IMA-AP in the Living Heart Human Model. IMA-AP acts nearly directly on peak-systolic AP (disease 26.1 mm $\rightarrow$ IMA-AP 50% $\rightarrow$ **15.9 mm**). Undeformed diastolic AP 34.4 mm is a different cardiac phase.

What remains for software methods is transparent **dose translation, provenance, held-out honesty, and exploratory ranking** under stated assumptions—not a claim of first CS-vs-AP comparison, not new FEA, and not a clinical recommendation engine.

2. Related work and positioning

Galili RSOS 2022 (doi:10.1098/rsos.211464) + Dryad supporting files provide the computational leakage/ROA/AP table we anchor to. Clinical series (ARTO/MAVERIC; Carillon TITAN II; REDUCE-FMR)

supply **directional / geometric context only**. This paper contributes research-software packaging, fold-wise response diagnostics, and an inspectable Layer-1 ranker, not a new high-fidelity FSI study.

3. Software architecture

```
configs/    surrogate_calibration.yaml, design_space.yaml, case YAML
models/     geometry, pathology, devices, response_model (fold-wise f_ROA / f_leak)
simulation/ phenomenological mechanics proxy (legacy alias run_fea_surrogate)
sph/        literature-calibrated leakage proxy (SPH-inspired; not a full SPH solver)
analysis/   ROA, jet, sweep, scenario_ranker, fit_response_model, paper tables/plots
tools/      import_galili_dryad.py, loo_evaluate.py (blend-off diagnostic), package_reports.py
data/       raw drop zone, fixtures, processed galili_cases.csv
results/output/cross_validation/ true fold-wise CV artifacts
```

Runnable entry points: `run_pipeline.py`, `python -m analysis.planner`, `python -m analysis.fit_response_model`, `python -m analysis.design_sweep`.

4. Methods

Pathology and geometry. Parametric papillary pathology (44% posterior passive). Undeformed diastole and peak systole are separate cardiac_phase records. AP diameter is a literature-defined geometry **input** (`ap_role: literature_mapping_input`); AP MAE is not reported as predictive accuracy.

Device kinematics. Galili mode interpolates published peak-systolic AP; clinical/planning mode uses assumption-prior η or preferred ΔAP . η_{CS} is **not** fitted from MAVERIC/ARTO. Default uncertainty samples η_{AP} and η_{CS} **independently** ($U(0.24, 0.36)$ and $U(0.44, 0.66)$).

Mechanics and leakage proxies. Algebraic coaptation-gap / strain / contact proxy scores. Leakage proxy is literature-calibrated at pathology. Physics regurg % $\neq$ clinical regurgitant volume. Synthetic contact-cluster ROA is visualization-only (`model_weight=1`, `cluster_weight=0`); Dryad `contact_fraction` enters the response model as a science feature.

Provenance. `tools/import_galili_dryad.py` downloads or documents manual drop; extracts Dryad `contact_fraction` and `n_sph_particles`. Scalar AP/ROA/leakage remain table-backed (not invented from unlabeled leaflet nodes).

Diagnostics vs true CV. `tools/loo_evaluate.py` is an **anchor-free / leave-one-case blend-off diagnostic** on the fixed rule-based surrogate (not fold-wise refit). `analysis/fit_response_model.py` performs **true fold-wise** fitting with held ROA/leak excluded from train arrays.

Scenario ranker. Minimizes leakage proxy; reports `n_total_points`, `n_device_candidates`, `n_feasible_device_candidates`, `p_feasible_device_candidates`. Pareto families:

pareto_global_common_objectives (min leak, max AP↓), pareto_ima_cs_family (adds LCx/NiTi), pareto_ima_ap_family. Missing LCx/NiTi are never filled with 1e9/0. Dual-suture factor is an explicit hypothesis. LCx prefers patient-measured baseline. The 14–20% AP band is a clinically contextualized **exploratory planning range** (anchor ~14–15%, ceiling 20%).

5. Results

5.1 Data provenance

Dryad archives yield deformed-contact and SPH coordinate files. Processed table data/processed/galili_cases.csv marks source_tier=dryad_derived for extracted contact_fraction and n_sph_particles. Published peak-systole AP, ROA, and leakage scalars remain table-backed fills — the importer does **not** extract independent AP, ROA, and leakage descriptors from unlabeled coordinates alone.

5.2 Calibration reproduction

Case	Leakage %	Peak-sys AP mm	ROA mm ²
pathology	5.26	26.1	172.8
ima_cs_22	0.29	24.8	48.2
ima_ap_50	0.08	15.9	27.3
ima_ap_70	0.13	12.4	46.1

Calibration IDs with blend ON are reproduction only.

5.3 Anchor-free diagnostic and true fold-wise CV (includes AP70 failure/improvement)

Rule-based blend-off diagnostic (tools/loo_evaluate.py, held-out IDs): historical pre-dampen baseline MAE ROA **50.18 mm²**, leak **1.445 pp**; IMA-AP 70% previously **140.3 / 5.585%** (abs err **94.17 / 5.455**). Current dampened rule-based held-out MAE $\approx$ **28.6 mm² / 0.66 pp**. Critically, the rule-based path **captures a qualitative non-monotonic tendency** (Galili AP50 leak 0.08% $\rightarrow$ AP70 0.13%) but **substantially overestimates AP70 leak magnitude** (pred $\approx$ 2.46% vs published 0.13%; abs err $\approx$ 2.33 pp) — this is **not** an unqualified claim that the surrogate “reproduces Galili non-monotonic behavior.”

True fold-wise response-model LOO (analysis/fit_response_model.py $\rightarrow$ results/output/cross_validation/): held-out-subset MAE ROA $\approx$ **6.4 mm²**, leak $\approx$ **0.25 pp**; AP70 abs err $\approx$ **6.6 mm² / 0.20 pp**. Internal engineering targets (ROA MAE < 25, leak MAE < 0.5, AP70 ROA < 25, AP70 leak < 0.5) are met on this seven-case table — **not** patient-level external validation. Full seven-fold MAE remains inflated by the pathology leave-out (large ROA scale gap). AP is literature geometry input (ap_prediction_metric_applicable=false).

Full held-out table (CS14 / CS18 / AP30 / AP70) — published vs rule-based blend-off vs fold-wise response model:

Case	Pub ROA	Rule ROA (abs err)	Fold ROA (abs err)	Pub leak %	Rule leak (abs err pp)	Fold leak (abs err pp)
ima_cs_14	56.7	24.0 (32.7)	69.6 (12.9)	0.52	0.65 (0.13)	0.60 (0.08)
ima_cs_18	55.3	17.6 (37.7)	53.4 (1.9)	0.41	0.34 (0.07)	0.42 (0.01)
ima_ap_30	51.1	14.9 (36.2)	47.0 (4.1)	0.16	0.28 (0.12)	0.86 (0.70)
ima_ap_70	46.1	53.9 (7.8)	39.5 (6.6)	0.13	2.46 (2.33)	0.33 (0.20)

AP70 row is reported front-and-center: fold-wise meets internal engineering screens; rule-based leak remains a prominent failure mode.

5.4 Exploratory scenario ranking (planning map, seed=42; assumption-driven)

- **n_total_points=36, n_device_candidates=35, n_feasible_device_candidates=30, p_feasible_device_candidates≈0.857**
- Under the **nominal dual-suture hypothesis**, best candidate ranked **IMA-AP dual 60%** ($\eta_{ap}=0.30 \rightarrow$ AP reduction **18.0%**, leakage proxy **~0.074%**, jet=central); leakage delta vs matched single ≈ -0.0018 pp. Ranking is assumption-sensitive (η sampling and dual factor).
- Best IMA-CS: **20%** bridge, CS-LCx **8.6 mm** at risk-screen edge
- Dual-suture jet rule: exploratory hypothesis only (Fig. 5 = factor sensitivity, not discovery)
- η uncertainty: independent η_{AP}/η_{CS} ; stability in scenario_ranking.json $\rightarrow$ uncertainty
- Pareto: global common objectives; CS/AP family frontiers without N/A fillers

5.5 Directionality-only clinical context

ARTO/MAVERIC: AP↓. Carillon TITAN II: ~15% AP context. REDUCE-FMR: regurg↓. Magnitudes not equated to physics %.

6. Discussion / limitations

Central advance: transparent exploratory screening software with corrected Galili peak-systole AP semantics, MAVERIC=ARTO mapping, Dryad-aware contact provenance, honest held-out/CV reporting (including AP70), and family-correct Pareto language. Limitations: algebraic proxies; Dryad CDN may require manual zip drop; seven discrete Galili cases are not patient-level external validation; η are priors; LCx/NiTi screens are literature/engineering screens not safety clearances; rule-based AP70 still overestimates leak magnitude relative to Galili.

7. Availability

- Code: <https://github.com/Coucou2016/fmr-ima-layer1-planner>
- License: MIT (LICENSE)
- Citation: CITATION.cff · references.bib
- Model card: MODEL_CARD.md · Data: DATA_PROVENANCE.md

8. Conclusion

A Layer-1 literature-anchored surrogate can rank exploratory IMA scenarios under explicit assumptions after correcting peak-systole Galili AP facts and MAVERIC device-class attribution, and after reporting true fold-wise response diagnostics alongside a renamed blend-off casewise diagnostic. Cite as methods/software for exploratory screening—not as clinical preoperative decision support.

Collab *notes:* docs/chatgpt_collab/20260913_p0_review_response.md,
docs/chatgpt_collab/20260914_plp2_implementation.md,
docs/chatgpt_collab/20260914_round2_major_revision.md,
docs/chatgpt_collab/20260915_r2_major_revision.md

Embedded SciencePlots figures

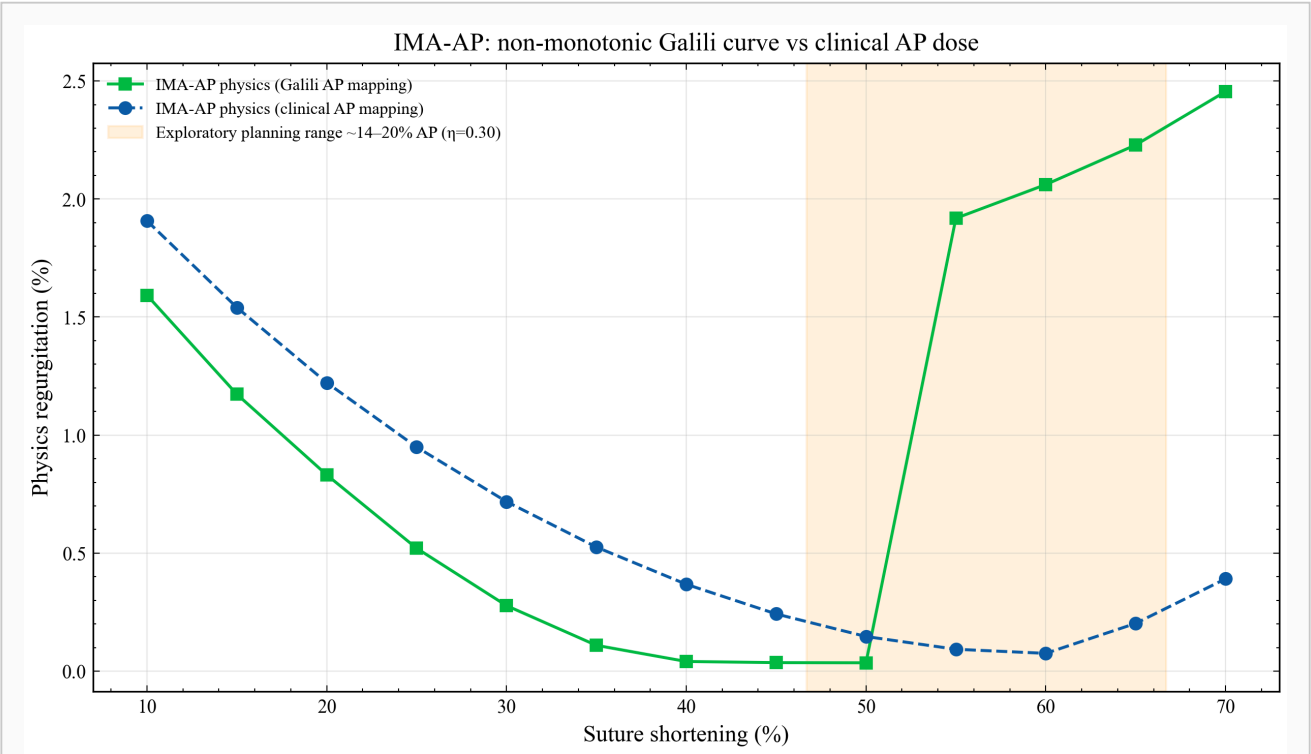


图 1. IMA-AP 物理反流随缝线缩短百分比的变化：Galili 映射 vs 临床映射

文件: fig1_ima_ap_nonmonotonic_clinical_window.png (SciencePlots + Times New Roman, dpi≥300; 坐标轴为英文标签, 释义见下文中文说明)

来龙去脉与读图说明: 本图回答“计算缝线缩短百分比能否直接当作临床前后径 (AP) 剂量?”横轴为 IMA-AP 缝线缩短 %; 纵轴为 Layer-1 泄漏代理 physics_regurgitation_pct (代数力学代理 + 文献校准泄漏代理; 不是超声反流容积分数)。

如何读: (1) Galili 映射复现发表峰缩期 AP: IMA-AP 50% → 峰缩期 AP=15.9 mm (相对疾病 26.1 mm 约 39% AP 缩减), 并非未变形舒张期 34.4 mm / 0% AP; 70% → 12.4 mm 并交界区泄漏升高。(2) 规划映射采用假设先验 $AP_reduction\% = \eta \times shortening\%$, 默认 $\eta_{ap}=0.30$ (假设, 非临床标定), 使 50% 缝线对应约 15% AP, 落入 ARTO/MAVERIC (IMA-AP 类, 不是 Carillon) 约 14–15% 对话窗口。绿色带为约 14–20% AP 窗口。

结论 (仅限本代理): 规划映射下窗口内总体可筛查; Galili 峰缩期映射下 70% 非单调恶化。勿将纵轴等同临床试验反流容积。

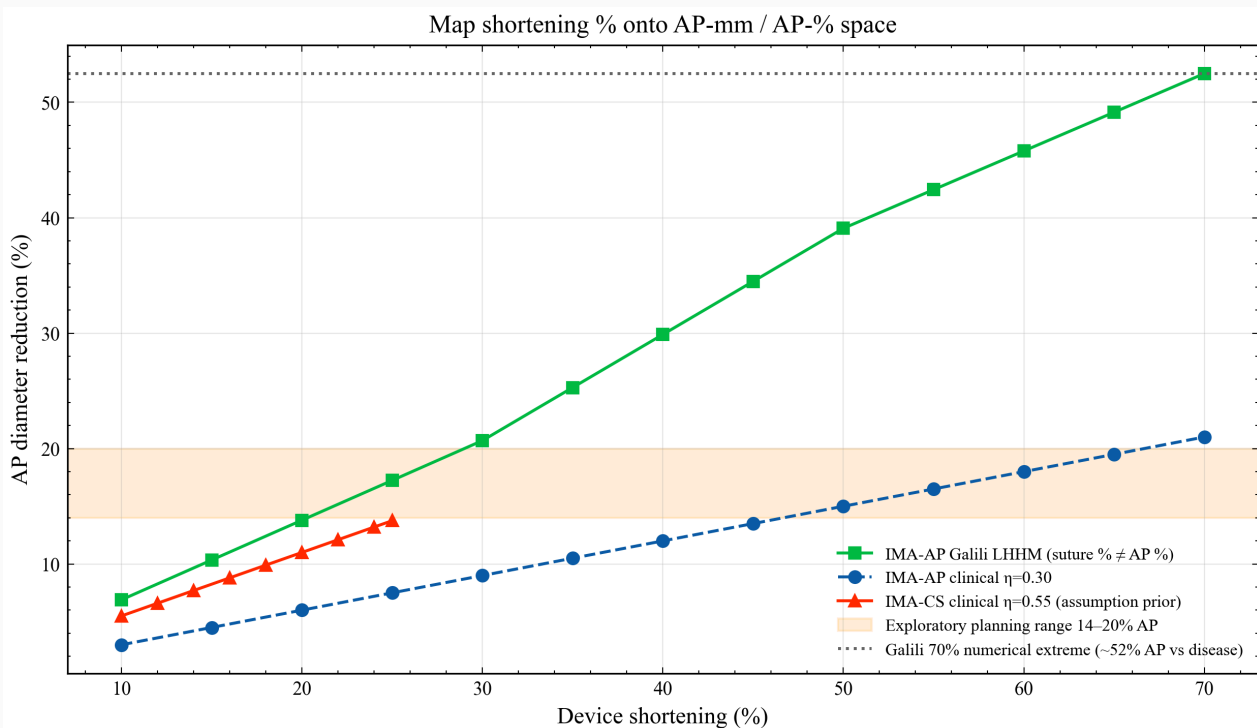


图 2. 装置缩短百分比到 AP 直径缩减百分比的剂量映射

文件: fig2_suture_vs_ap_reduction.png (SciencePlots + Times New Roman, dpi≥300; 坐标轴为英文标签, 释义见下文中文说明)

来龙去脉与读图说明: 横轴为缩短 %; 纵轴为 AP 缩减 %。轨迹对照: Galili 峰缩期 (IMA-AP 近直接缩 AP)、

规划 IMA-AP (假设 $\eta_{ap}=0.30$)、规划 IMA-CS (假设 $\eta_{cs}=0.55$, 非 MAVERIC 拟合; Carillon/TITAN II ~15% 仅语境)。

MAVERIC=ARTO (IMA-AP 类)。水平带 14–20% 为探索性规划窗口。

如何读: 同一缩短 % 在不同映射下 AP 缩减截然不同。 η 为假设先验, 不是成像–力学辨识, 也不是“临床标定常数”。

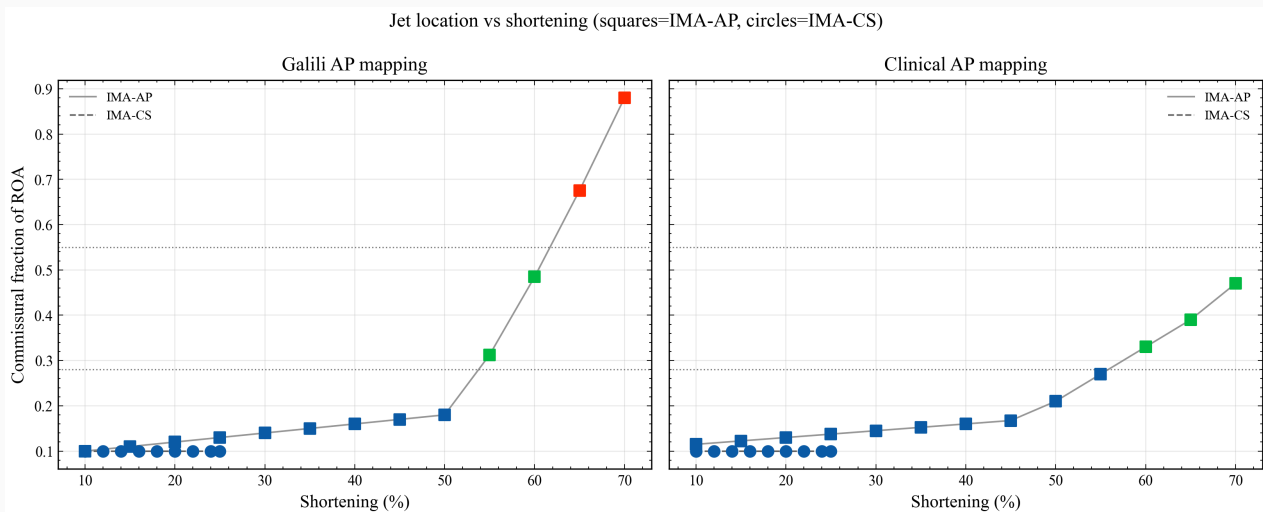


图 3. 射流位置与交界区 ROA 分数随缩短变化（Galili vs 临床）

文件: fig3_jet_location.png (SciencePlots + Times New Roman, dpi≥300; 坐标轴为英文标签, 释义见下文中文说明)

来龙去脉与读图说明: 反流“量”之外, 筛查还关心泄漏机制位置。本代理将 ROA (regurgitant orifice area,

反流口面积, 单位 mm^2 , 由对合间隙等代理量估计) 拆为中央份额与交界区份额, 并分类为 central / commissural / mixed。横轴为缩短 %; 纵轴为交界区 ROA 分数; 方块=IMA-AP, 圆点=IMA-CS; 左右面板对照

Galili 与临床映射。水平虚线为分类阈值示意。双缝线 $\times 0.5$ 为显式假设参数, 不是独立“发现”。

如何读: 交界区分数升高意味着泄漏更偏交界区; 分类从 central 变为 mixed/commissural 表示机制标签切换。

该标签是代理口上的机制草图, 不是超声 PISA (proximal isovelocity surface area) 或多普勒诊断。

seed-42 假设下最优候选双缝线 60% 的机制来源可在本图与图 5 对照中追溯: 同 AP 下交界份额更低、jet 保持 central。

结论: 在临床映射下, IMA-AP 高剂量单缝线更易出现 mixed; IMA-CS 在可行桥缩短范围内多保持 central。

结合图 5, 双缝线可在匹配 AP 缩减下压低交界区分数 (假设情景)。

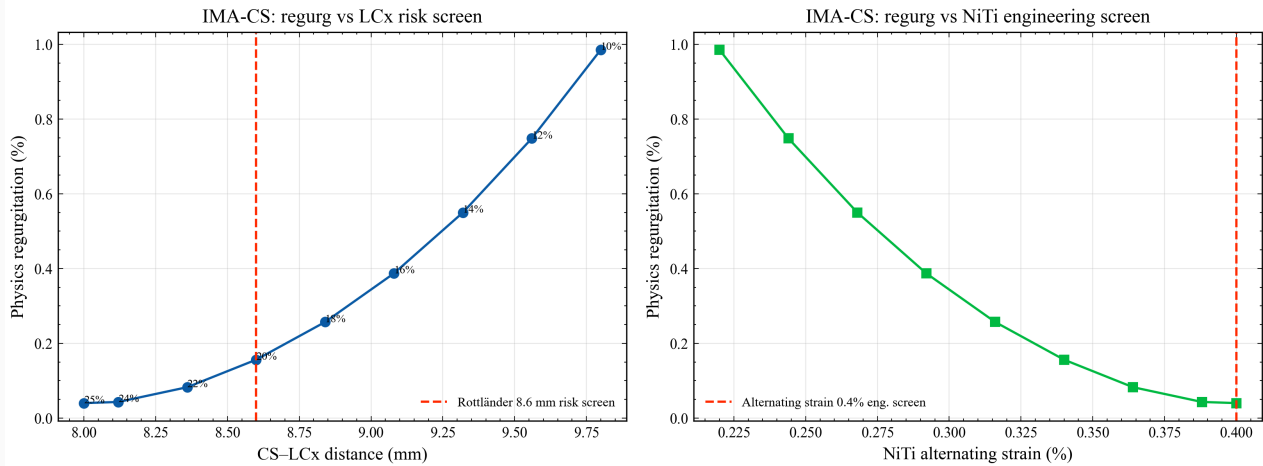


图 4. IMA-CS 临床映射：反流–筛查 Pareto（CS–LCx 与 NiTi 交变应变）

文件：fig4_pareto_lcx_strain.png（SciencePlots + Times New Roman，dpi≥300；坐标轴为英文标签，释义见下文中文说明）

来龙去脉与读图说明：IMA-CS 在降低反流的同时可能压缩 CS–LCx 距离（coronary sinus–left circumflex artery，

冠状窦与左回旋支间距）。文献筛查边界取远端着陆区 CS–LCx ≥ 8.6 mm（Rottländer 等；注意：<8.6 mm

预测妥协 ≠ ≥8.6 mm 已证明安全）。同时对 NiTi（镍钛合金桥）交变应变设置 <0.4% 的工程筛查上限。默认示意解剖基线 CS–LCx=11.0 mm，可替换为患者 CT。标题用“筛查”而非“安全证明”。

如何读左右面板：横轴多为筛查指标（CS–LCx 或交变应变），纵轴为 physics 反流；点随桥缩短移动，越接近约束边界，可行域越窄。22% 及以上在默认解剖上常因 CS–LCx 违约束而不可行。

结论：默认解剖上可行最优贴近 CS–LCx=8.6 mm 的桥缩短 20%。该结果是规划器筛查信号，不是患者级安全证明。

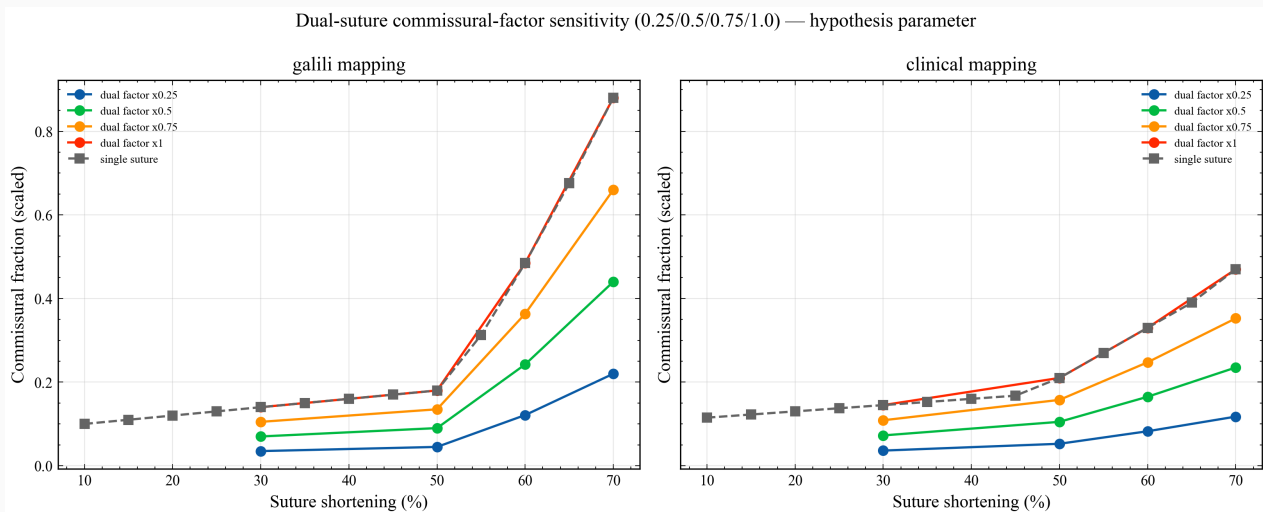


图 5. 双缝线交界因子敏感性（0.25/0.5/0.75/1.0）——假设参数，非 **Innovation D** 发现

文件: fig5_dual_vs_single_suture.png (SciencePlots + Times New Roman, dpi≥300; 坐标轴为英文标签, 释义见下文中文说明)

来龙去脉与读图说明: 本图展示双缝线交界因子 (0.25/0.5/0.75/1.0) 敏感性, 不是“**Innovation D discovery**”。

名义因子 $\times 0.5$ 为显式假设参数; 排序对因子选择敏感。勿将 physics 泄漏差 (如 dual60 vs single ~ 0.0018 pp) 写成临床反流容积改善。